Data Processing Facilities

[Source: Page 97 to 98, CISSP for Dummies]

Data processing facilities are so vital to businesses today that a lot of emphasis is placed on them. Generally, this comes down to these variables: where and how the business will continue to sustain its data processing functions.

Because data centers are so expensive and time-consuming to build, better business sense dictates having an alternate processing site available. The types of sites are:

Cold site:
A cold site is basically an empty computer room with environmental facilities (UPS; heating, ventilation, and air conditioning [HVAC]; and so on) but no computing equipment.

Warm site:
A warm site is basically a cold site, but with computers and communications links already in place. In order to take over production operations, you must load the computers with application software and business data.

Hot site:
Indisputably the most expensive option, you equip a hot site with the same computers as the production system, with application changes, operating system changes, and even patches kept in sync with their live production-system counterparts. You even keep business data up-to-date at the hot site by using some sort of mirroring or transaction replication. Hot sites may be cloud-based or in a co-location center.

Reciprocal site:
Your organization and another organization sign a reciprocal agreement in which you both pledge the availability of your organization's data center in the event of a disaster. Back in the day, when data centers were rare, many organizations made this sort of arrangement, but it's fallen out of favor in recent years.

Multiple data centers:
Larger organizations can consider the option of running daily operations out of two or more regional data centers that are hundreds (or more) of miles apart. The advantage of this arrangement is that the organization doesn't have to make arrangements with outside vendors for hot/warm/cold sites, and the organization's staff is already onsite and familiar with business and computer operations.

Cloud site:
Organizations with primary information processing in the cloud are likely to employ cloud assets in multiple regions, and possibly with more than one cloud provider. Many organizations with primary processing on-premises employ hybrid cloud infrastructure for disaster recovery purposes — this is a common way for companies to ease their way into the cloud. Depending on the degree of readiness

required, a cloud site can be as ready as a hot site, a warm site, or a cold site, as determined by the resources devoted to keeping the cloud site ready for production operations.

Table 3-1 compares these options side by side

TABLE 3-1 Data Processing Continuity Planning Site Comparison

Feature	Hot Site	Warm Site	Cold Site	Multiple Data Centers	Cloud Site
Cost	Highest	Medium	Low	No additional	Variable
Computer-equipped	Yes	Yes	No	Yes	Yes
Connectivity-equipped	Yes	Yes	No	Yes	Yes
Data-equipped	Yes	No	No	Yes	Variable
Staffed	Yes	No	No	Yes	No
Typical lead time to readiness	Minutes to hours	Hours to days	Days to weeks	Minutes to hours or longer	Minutes to hours